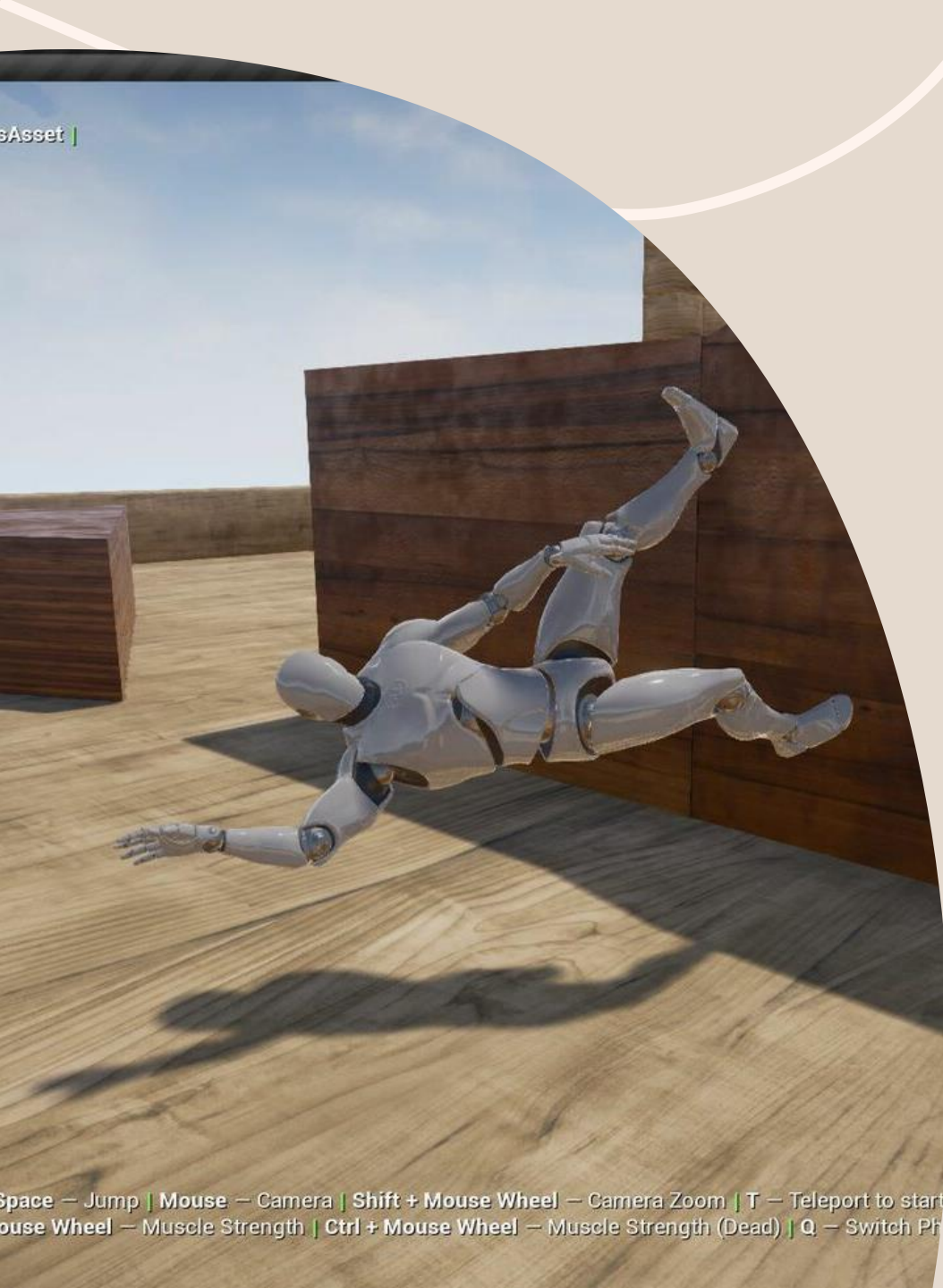




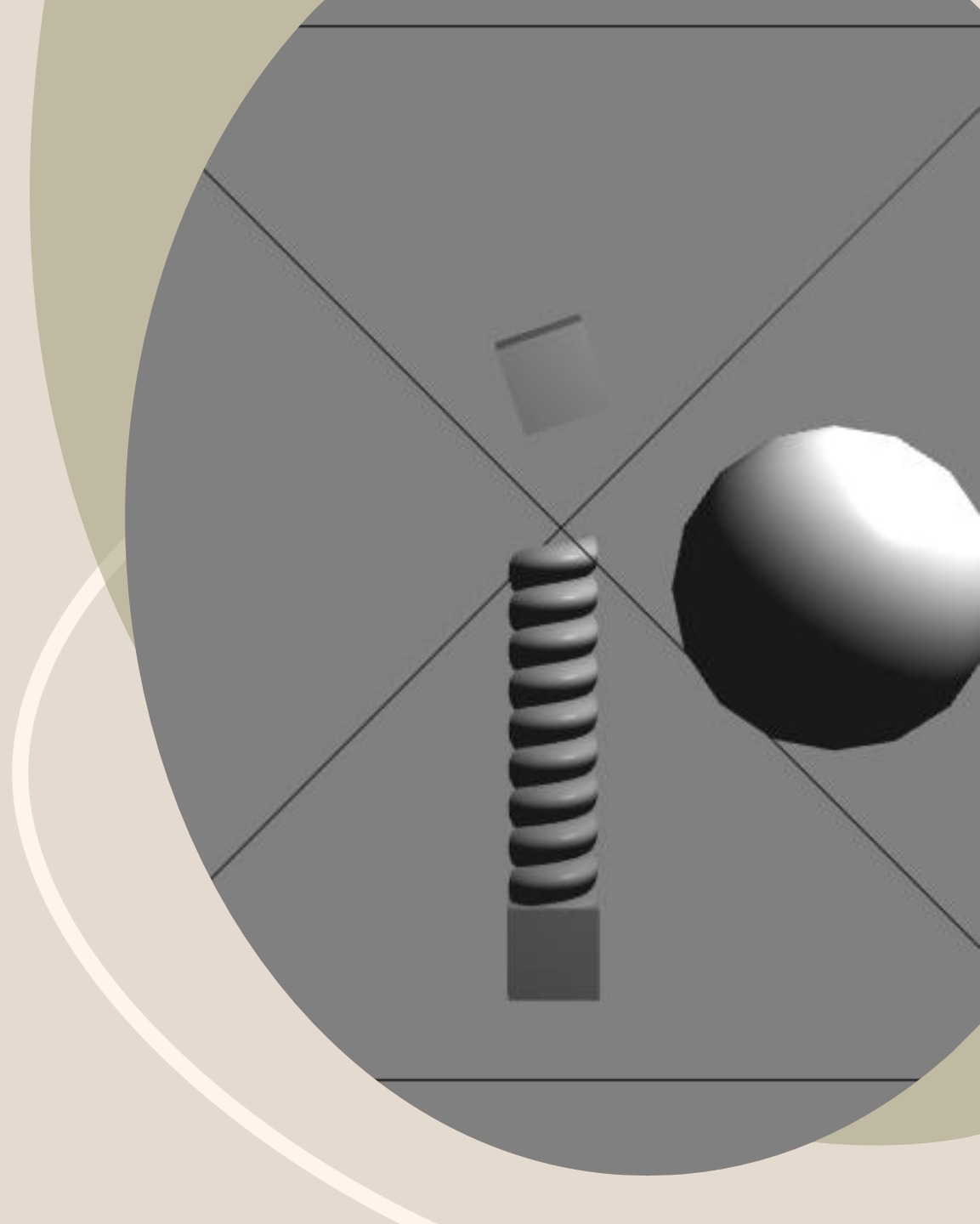
ΓΡΑΦΙΚΑ ΚΑΙ ΕΙΚΟΝΙΚΗ ΠΡΑΓΜΑΤΙΚΟΤΗΤΑ

ΕΡΓΑΣΤΗΡΙΟ 7: PHYSICS

Space — Jump | Mouse — Camera | Shift + Mouse Wheel — Camera Zoom | T — Teleport to start
Mouse Wheel — Muscle Strength | Ctrl + Mouse Wheel — Muscle Strength (Dead) | Q — Switch Ph

Physics

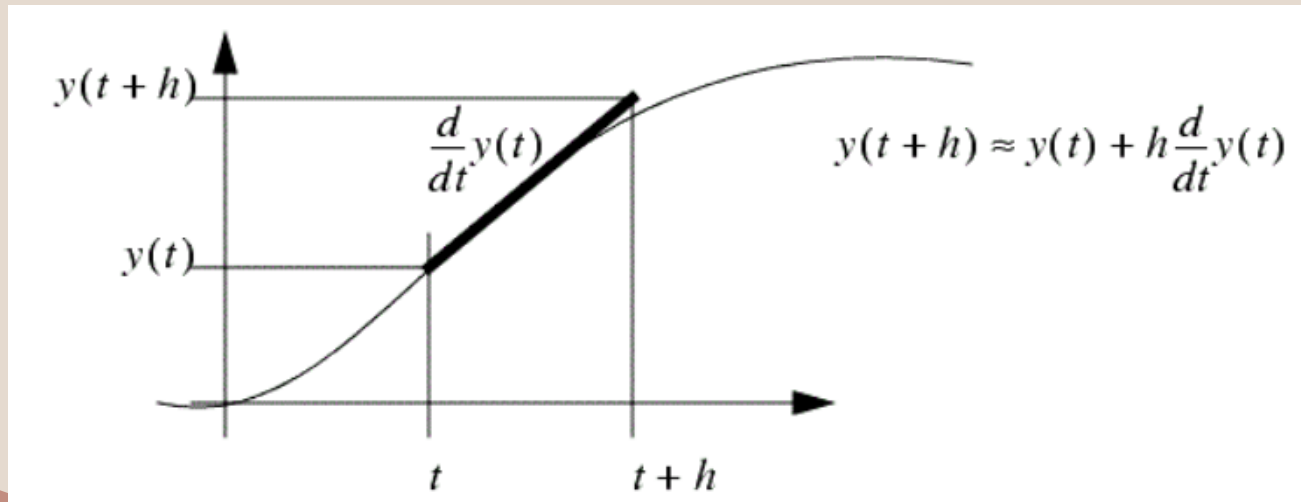
- Linear Motion
- Collisions (elastic)
- Spring oscillations



Numerical Integration: Euler's method

Given an initial state, I want to predict future states w.r.t. time.

From Taylor's expansion: $y(t + h) = y(t) + h \frac{dy}{dt} + \frac{h^2}{2!} \frac{d^2y}{dt^2} + \dots$



Approximating by taking the first two terms and a step of our choice.

Larger step = larger error. $\frac{\text{error}}{\text{step}} O(h^2)$

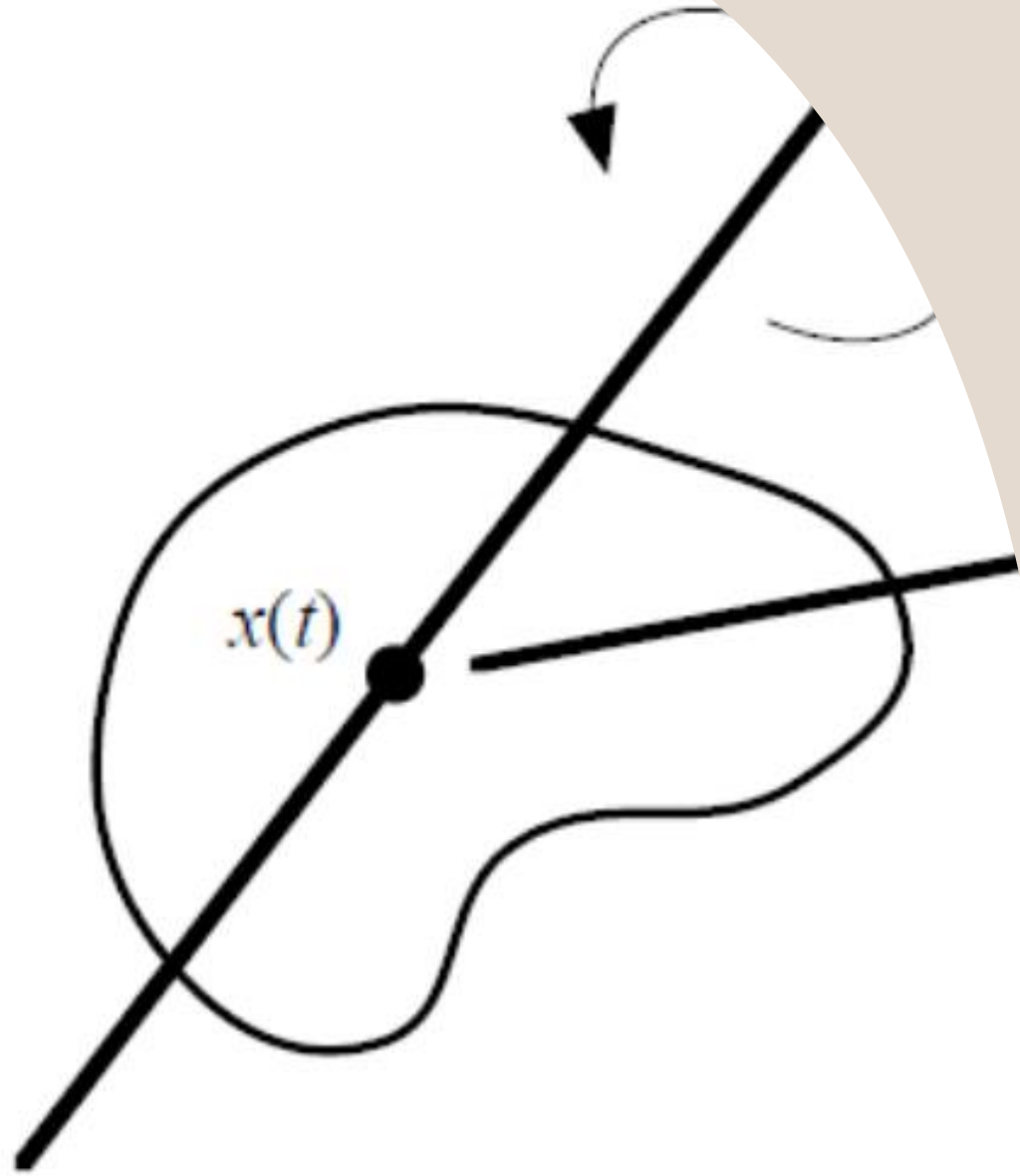
Rigid Body

Q: WHAT VALUES DO I NEED TO
CAPTURE EVERY FRAME?

For Drawing

For Handling Physics:

Hint: need to be
able to at least
guess the 1st
derivative



Rigid Body

Q: WHAT VALUES DO I NEED TO CAPTURE EVERY FRAME?

For Drawing

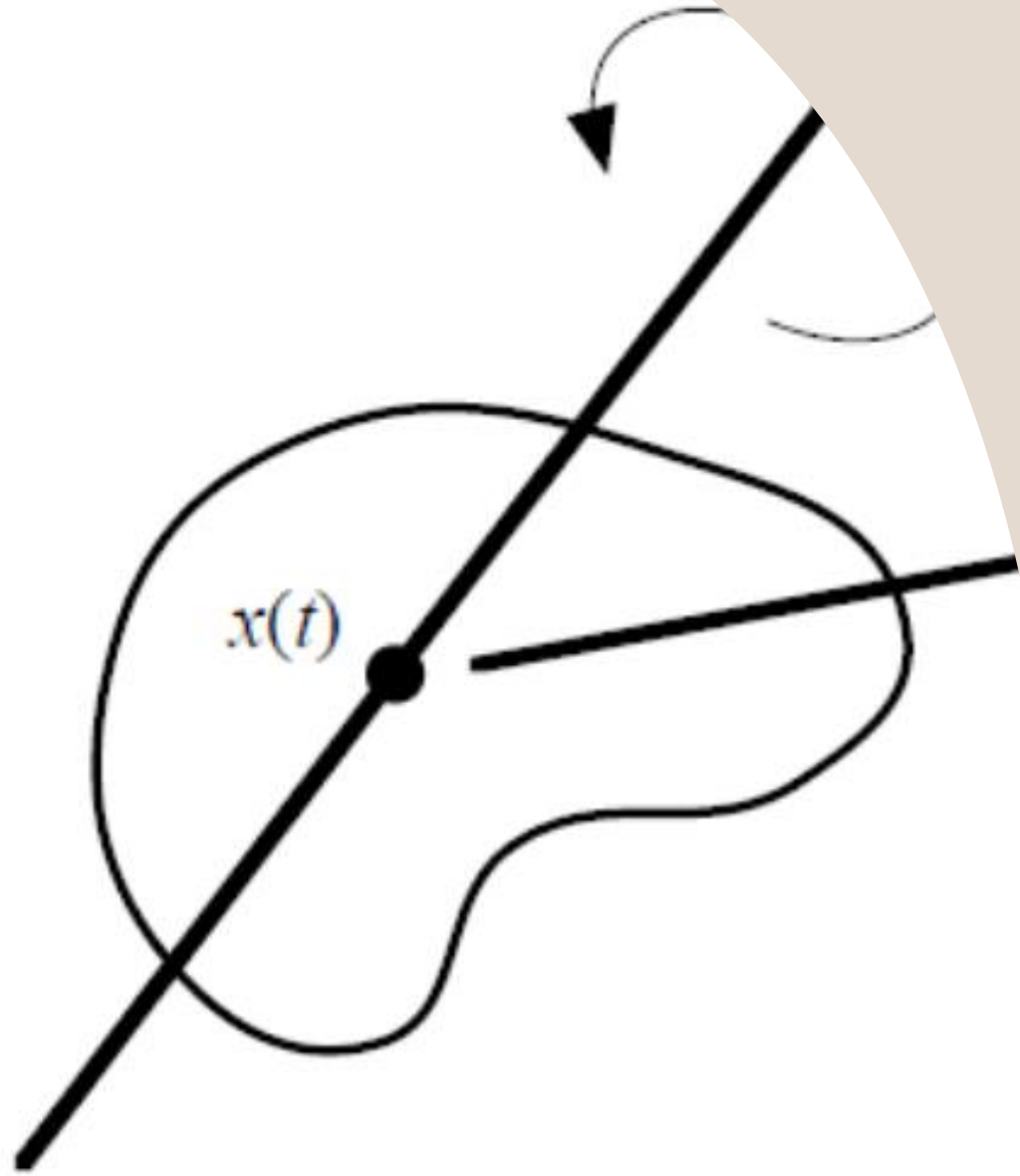
- $x(t)$
- $R(t)$

For Handling Physics:

- $P(t) = mv(t)$
- $L(t) = I(t)\omega(t)$

$$y(t) = \begin{bmatrix} x(t) \\ R(t) \\ P(t) \\ L(t) \end{bmatrix}$$

$$\frac{dy(t)}{dt} = \begin{bmatrix} v(t) \\ \omega(t)R(t) \\ F(t) \\ \tau(t) \end{bmatrix}$$

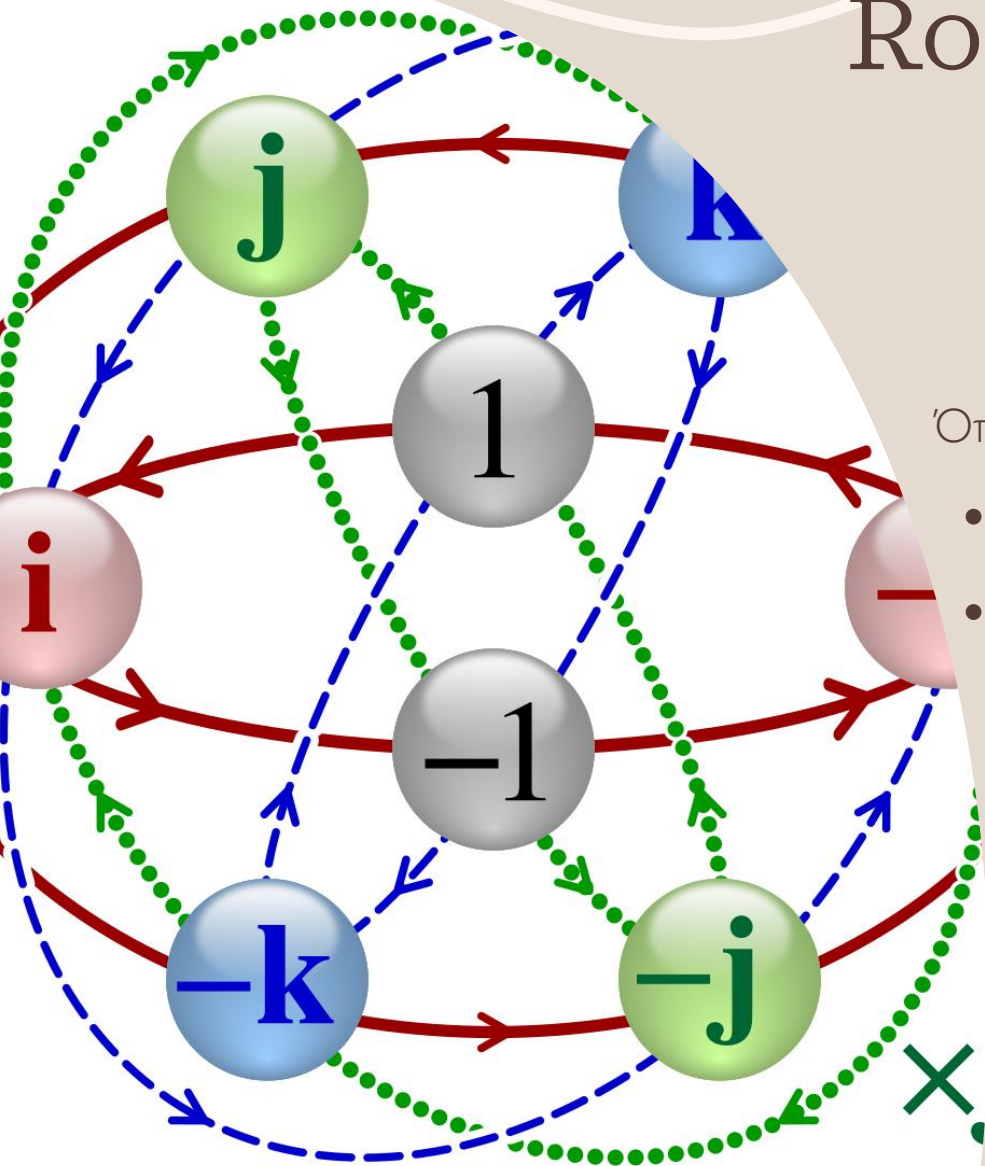


Task 1: Cube (Newton's 1st Law)

Task 1a: change the parameters of the cube and observe its behavior

Task 1b: use rotation matrix to encode the orientation of a rigid body
Comment the definition (`#define USE_QUATERNIONS`) in `RigidBody.h`.
What do you observe?

Quaternions vs Rotation Matrices



Quaternions

$$y = \alpha + \beta i + \gamma j + \delta k$$

Όπου:

- $i^2 = j^2 = k^2 = -1$
- $ij = k, jk = i, ki = j$

Quaternions describe 4D coordinates, but they can express 3D rotations when normalized into unit quaternions $\|q\| = 1$

- More numerically stable (Rotation Matrices require orthogonality)
- Cheaper (only 4 floats instead of 9)
- Less intuitive
- Discontinuous ($q = -q$)

Task 2: Sphere (Collisions)

Task 2a: change the initial velocity and position, mass and radius (lab.cpp)

Task 2b: define the velocity of the sphere after the collision (Collision.cpp)

Task 2c: model the force due to gravity (lab.cpp)

Task 2d: calculate the total energy and comment on the previous task

The kinetic energy of a rigid body can be calculated as follows:

$$KE = \frac{1}{2}mu^T u + \frac{1}{2}\omega^T I \omega$$

And the potential energy (of a conservative force):

$$U = - \int_{y_0}^y F(y) dx$$

Task 2e: Runge-Kutta (4th order)

- $y_0 = y(t)$

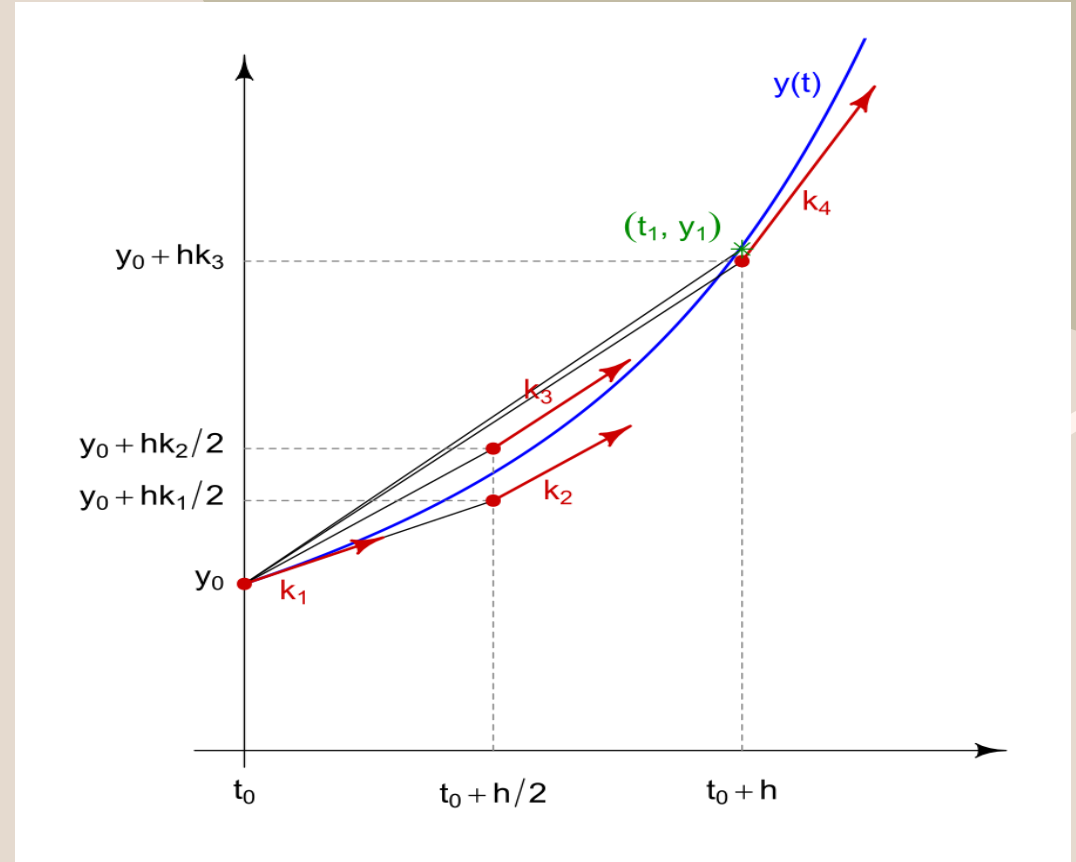
$$k_1 = \frac{dy_0}{dt} = f(y_0, dt) \quad y_1 = y_0 + \frac{h}{2}k_1$$

$$k_2 = f(y_1, dt + \frac{h}{2}) \quad y_2 = y_0 + \frac{h}{2}k_2$$

$$k_3 = f(y_2, dt + \frac{h}{2}) \quad y_3 = y_0 + \frac{h}{2}k_3$$

$$k_4 = f(y_3, dt + h)$$

$$y(t+h) = y_4 = y_0 + h \frac{k_1 + 2k_2 + 2k_3 + k_4}{6} \quad O(h^5)$$



Task: Change dt to **0.001f** and observe the total energy, then change the numerical integration method to **Runge-Kutta 4th order** (in RigidBody.cpp)

Q1: Why did Euler always overshoot instead of undershooting?

Q2: Observe and explain the behavior of the total energy now!

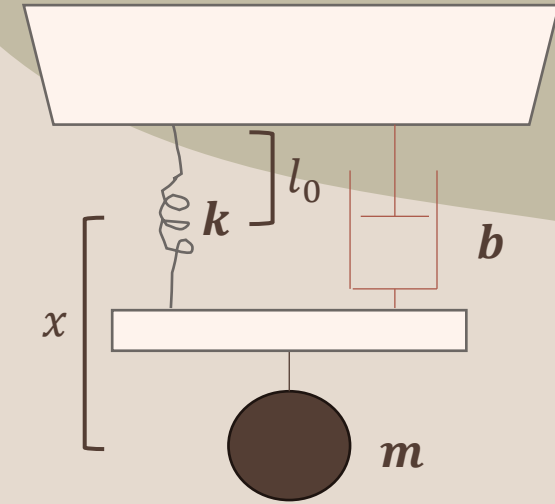
Task 3: Damped Spring

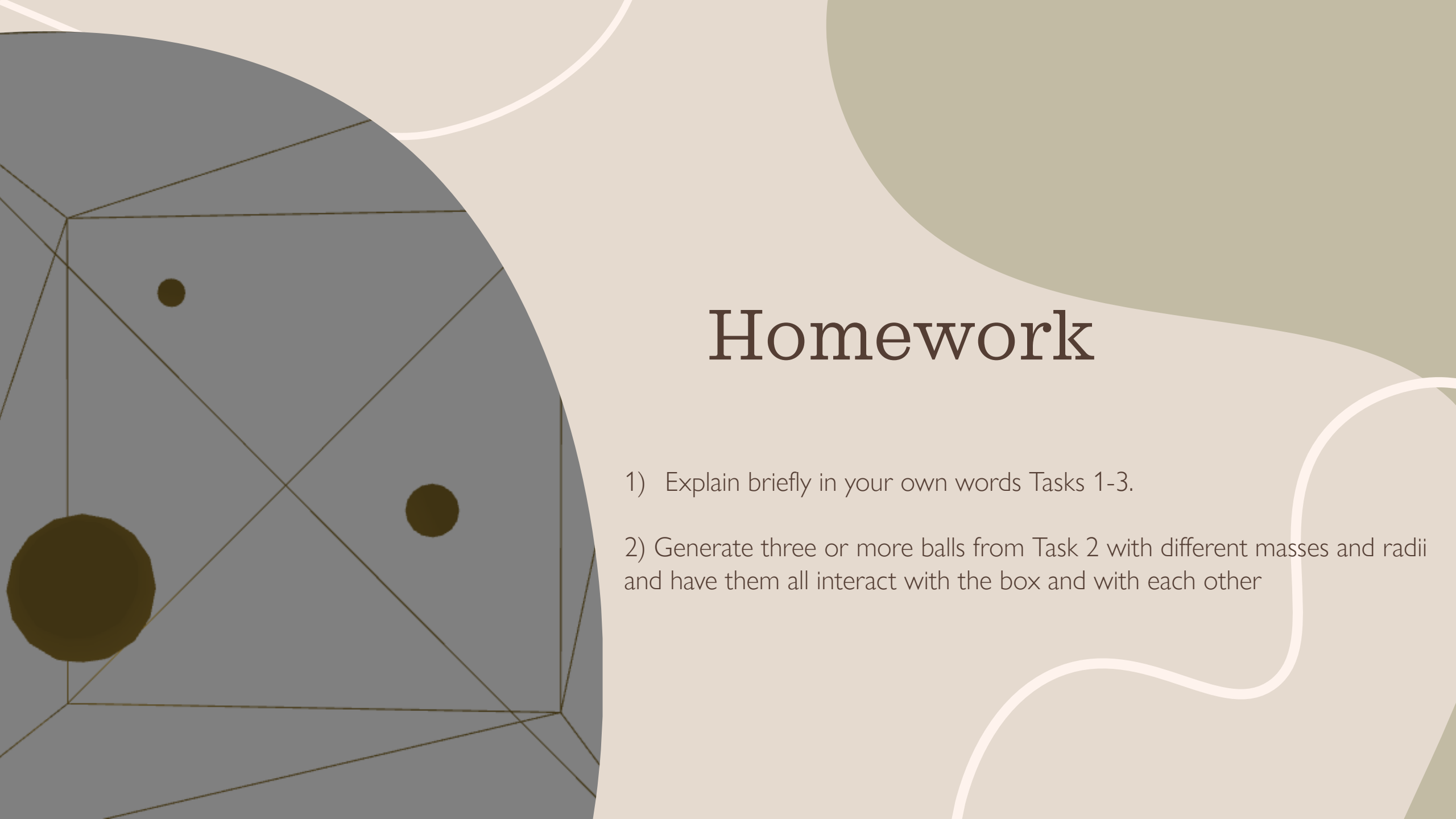
Task 3a: model a mass-spring-damper system
(uncomment) (lab.cpp)

Task 3b: change the initial parameters of the system

Task 3c: model the forces due to gravity and spring

Task 3c: model the (linear) damping force



The background features a light beige color with large, soft-edged organic shapes in a muted olive green and a darker taupe. On the left side, a portion of a sphere is visible, rendered in a dark grey-blue. This sphere has a white wireframe grid of lines and contains three solid black circles of varying sizes, suggesting a celestial or scientific theme.

Homework

- 1) Explain briefly in your own words Tasks 1-3.
- 2) Generate three or more balls from Task 2 with different masses and radii and have them all interact with the box and with each other